Project Progress Report

Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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1 Project Overview

This project provides a systematic numerical characterization of hyperuniform point patterns in one dimension. Two complementary metrics are computed for each pattern:

- **Hyperuniformity exponent** α: characterizes how $S(k) \sim |k|^\alpha$ as $k \to 0$.
- **Surface-area coefficient** $\bar{\Lambda}$: measures residual disorder within the strongest (Class I) patterns; $\bar{\Lambda} = 1/6$ is the global minimum (integer lattice).

Three universality classes arise from the large-R behavior of the number variance $\sigma^2(R)$:

Class	Condition	Variance	$\bar{\Lambda}$
I	$\alpha > 1$	bounded	finite
II	$\alpha = 1$	$\sim \Lambda_{II} \ln R$	diverges
III	$0 < \alpha < 1$	$\sim \Lambda_{III} R^{1-\alpha}$	diverges

The project is organized by "weeks," each corresponding to a checkpoint meeting with Prof. Torquato (not necessarily one calendar week). Three such meetings have taken place to date, yielding 27 characterized patterns and a Junior Paper (§1–§3 complete, 9 pages; results sections pending). A fourth meeting cycle is now underway.

2 Literature Review

This section surveys the primary literature that motivates, informs, and contextualizes the project. Twelve papers are reviewed, organized thematically.

2.1 Foundations of Hyperuniformity

Torquato & Stillinger (2003). The foundational paper defines **hyperuniformity** as the condition $S(\mathbf{k} \to \mathbf{0}) = 0$, where $S(\mathbf{k})$ is the structure factor. For a d-dimensional point pattern the

number variance $\sigma^2(R) \equiv \langle N(R)^2\rangle - \langle N(R)\rangle^2$ grows at most as the surface area R^{d-1} for large R, in contrast to the R^d growth of a Poisson process. Three universality classes are introduced: Class I ($\alpha > 1$, bounded $\bar{\Lambda}$), Class II ($\alpha = 1$, $\sigma^2 \sim \Lambda_{II} \ln R$), Class III ($0 < \alpha < 1$, $\sigma^2 \sim \Lambda_{III} R^{1-\alpha}$).

The **surface-area coefficient**

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{\sigma^2(R)}{2\rho R^{d-1} A_1(R)}$$

(with $A_1(R)$ the window surface area) serves as an order metric for Class I patterns. For $d = 1$ the integer lattice achieves $\bar{\Lambda}_{\min} = 1/6$, derived from the exact formula $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ is the fractional part of R. In higher dimensions, Bravais lattices minimize $\bar{\Lambda}$ among periodic patterns. The paper establishes the conceptual framework adopted throughout this project.

Zachary & Torquato (2009). This paper extends hyperuniformity to **two-phase heterogeneous media**. In this setting the relevant quantity is the **spectral density** $\tilde{\chi}_V(\mathbf{k})$ — the Fourier transform of the two-point correlation of the indicator function $I(\mathbf{x})$ for phase 2. Hyperuniformity now requires $\tilde{\chi}_V(\mathbf{k} \to \mathbf{0}) = 0$, in direct analogy with $S(\mathbf{k} \to \mathbf{0}) = 0$ for point patterns.

A central technical contribution is the **theta-function (Bragg-sum) method** for computing $\bar{\Lambda}$ analytically from the Fourier-space structure of the pattern: Bragg peaks at reciprocal-lattice vectors $\mathbf{k}_j$ contribute to $\sigma^2(R)$ through overlap integrals of the window function. This method is applied to tabulate $\bar{\Lambda}$ for $d = 1, 2, 3$ Bravais lattices and one-dimensional quasicrystals. The Fibonacci chain result $\bar{\Lambda} = 0.20110$ (Table 1 of that paper) serves as the primary literature benchmark for this project's numerical validation.

2.2 Substitution Tilings: Classification

Öguz, Socolar, Steinhardt, Torquato (2019). This paper provides a comprehensive classification of hyperuniformity and **anti-hyperuniformity** in all one-dimensional substitution tilings. Given a substitution matrix $\mathbf{M}$ with Perron–Frobenius eigenvalue λ_1 and second-largest eigenvalue λ_2, the **eigenvalue formula**

$$\alpha = 1 - \frac{2\ln|\lambda_2|}{\ln \lambda_1}$$

expresses the hyperuniformity exponent entirely in terms of the matrix spectrum. The exponent α fills $[-1, 3]$ densely over the space of all valid substitution matrices, including anti-hyperuniform cases ($\alpha < 0$) where the structure factor *diverges* at the origin.

The Pisot–Vijayaraghavan (PV) property — $|\lambda_i| < 1$ for all $i \geq 2$ — is shown to be necessary and sufficient for the existence of **Bragg peaks** (non-zero weight in the diffraction measure at isolated k-values). Non-Pisot substitutions lack Bragg peaks entirely, and their diffraction measure is purely continuous; $Z(k)$ then decays faster than any power law, placing them outside the α-classification scheme. Concrete results: Fibonacci $\alpha = 3$ (Class I), Period-Doubling $\alpha = 1$ (Class II). These two cases are used throughout this project.

Bombieri & Taylor (1986). Although no PDF is available, this paper is a key reference: the authors studied the *cubic* 3-letter substitution $a \to aac$, $b \to ac$, $c \to b$ (equivalently encoded by a 3×3 integer matrix) and showed that it produces a well-defined diffraction pattern with Bragg peaks. With $\lambda_1 \approx 2.247$ and $|\lambda_2| \approx 0.802$ the eigenvalue formula gives $\alpha = 1.545$, the **first known example filling the gap** $1 < \alpha < 2$. This chain was reproduced in the present project with $\bar{\Lambda} = 0.377$.

2.3 Quasicrystals and Projection Methods

Shechtman, Blech, Gratias, Cahn (1984). The experimental discovery of an Al-Mn alloy with icosahedral diffraction symmetry — forbidden by classical crystallography — launched the field of quasicrystals. Bragg peaks appear at irrational wavevector ratios, implying long-range order without periodicity. This motivates the theoretical framework of *cut-and-project* (projection) methods in which a higher-dimensional periodic lattice is projected onto a lower-dimensional physical space along an irrational direction.

Öguz, Macé, Torquato (2017). This paper analyzes hyperuniformity of projection-method quasicrystals analytically. For the ideal Fibonacci strip (projection window aligned with the golden ratio), the structure factor satisfies $Z(k) \sim k^4$, giving $\alpha = 3$ (Class I). A nonideal strip (irrational window boundary) degrades to $\alpha = 1$ (logarithmic variance, Class II): the rational approximants to the boundary introduce quasi-periodic modulations that slow the decay of $S(k)$.

The paper derives an **analytical upper bound** $\sigma^2(R) \leq 1/4$ for the Fibonacci chain, consistent with the numerical Fibonacci $\bar{\Lambda} = 0.201 < 1/4$ found in this project (and with $\bar{\Lambda}_{\text{Silver}} = 1/4$ as a candidate upper-bound saturation for $n = 2$). The two-point correlation $g_2(r)$ is shown to be piecewise quadratic with inflection points at the Fibonacci spacings 1 and $\tau = (1 + \sqrt{5})/2$.

2.4 Stealthy Hyperuniform Ground States

Batten, Stillinger, Torquato (2008). This paper introduced the concept of **collective-coordinate optimization**: given a target function $\Phi = \sum_{|\mathbf{k}| \leq K} S(\mathbf{k})$, gradient descent on particle positions drives $S(\mathbf{k}) \to 0$ for $|k| \leq K$, engineering a **stealthy** configuration. The resulting patterns are highly degenerate ground states (many configurations satisfy the stealth condition) and are generally *disordered*. Special cases include super-ideal gas variants and equi-luminous structures. A key insight: stealth can be achieved with particles that appear to have no macroscopic order, yet exhibit complete isotropy and hidden long-range correlations. This paper predates the thermodynamic ensemble framework of Torquato et al. (2015) and provides the algorithmic foundation for all stealthy ensemble generation.

Torquato, Zhang, Stillinger (2015) — *Physical Review X*. This paper develops the **exact ensemble theory** for stealthy hyperuniform disordered ground states. The control parameter is

$$\chi = \frac{M(K)}{d(N-1)},$$

where $M(K)$ is the number of independently constrained wavevectors within a ball of radius K, N is the particle number, and d the dimension. For $\chi < 1/2$ the ground-state manifold is *disordered* (positive entropy); for $\chi > 1/2$ only periodic configurations satisfy the constraints. All thermodynamic quantities — isothermal compressibility κ_T, pair correlations $g_2(r)$, structure factor $S(k)$ — are derived analytically in the ensemble limit. The paper also derives the relation $K^* = 4\pi\rho\chi$ (for $d = 1$) between the exclusion radius K^* and the control parameter χ.

In $d = 1$ the paper's Table V provides reference values of $\bar{\Lambda}$ for stealthy configurations, which are well approximated by $\bar{\Lambda} \approx 1/(\pi^2\chi)$ — a formula derived independently in this project and validated against the collaborator's ensemble data.

2.5 Perturbed Lattices and Cloaking

Klatt, Kim, Torquato (2020). This paper introduces and analyzes the **uncorrelated random lattice (URL)**: each site of the integer lattice $\mathbb{Z}^d$ is displaced independently and uniformly within

a hypercube $[-a/2, a/2]^d$. In one dimension the **exact formula**

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6}$$

is derived analytically (for $a \leq 1$) and extended via periodicity to all $a > 0$. Crucially, the URL always has $\alpha = 2$ regardless of a: the uniform displacement smears the lattice Bragg peaks into a continuous background, but the origin $k = 0$ retains the same power-law suppression $S(k) \sim k^2$.

The central discovery is **cloaking**: when the displacement amplitude a equals an integer (1, 2, 3, ...), the form factor $\tilde{f}(k) = \sin(\pi k a)/(\pi k a)$ vanishes at every non-zero reciprocal lattice vector, completely eliminating the Bragg peaks from the diffraction pattern. At $a = 1$: $\bar{\Lambda} = 1/3$, which this project identifies as the *exact limit* of the metallic-mean sequence $\bar{\Lambda}(\mu_n) \to 1/3$ as $n \to \infty$. The τ order metric (based on pair correlations) diverges between cloaking events, providing a complementary characterization of the structural transition.

2.6 Diffusion Spreadability

Torquato (2021) — *Physical Review E.* This paper introduces **diffusion spreadability** $\mathcal{S}(t)$ as a dynamic probe of microstructure. Consider a two-phase medium (phases 1 and 2) with a solute initially confined to phase 1; $\mathcal{S}(t)$ measures the fraction of solute transferred to phase 2 by time t. For hyperuniform media with exponent α the long-time behavior is

$$E(t) \equiv \mathcal{S}(\infty) - \mathcal{S}(t) \ \sim \ t^{-(d+\alpha)/2},$$

while stealthy media show *exponential* decay $E(t) \sim e^{-ct}$. At short times $\mathcal{S}(t) \sim t^{1/2}$ in all cases, providing access to the **specific surface** s (surface area per unit volume of the interface). Applications include NMR/MRI characterization of porous media, where the diffusion signal encodes structural information across length scales. The 1D formula $E(t) \sim t^{-(1+\alpha)/2}$ is used in this project to extract α via a log-log regression of the excess spreadability.

Wang & Torquato (2022) — *Physical Review Applied.* This paper applies spreadability to a broad set of 2D and 3D model microstructures: URL packings, stealthy ground states, Debye random media (nonhyperuniform), Bravais lattices, and real porous media (Fontainebleau sandstone). A practical algorithm is presented to extract α from the slope of $\ln E(t)$ versus $\ln t$ in the asymptotic regime. For one-dimensional structures the paper gives $\phi_2 = 0.35$ for the Fibonacci chain (volume fraction of phase 2 in the two-phase embedding), which is used throughout this project for all 1D spreadability calculations.

Hitin-Bialus, Maher, Steinhardt, Torquato (2024). This paper achieves high-precision extraction of α for quasiperiodic tilings via spreadability. By exploiting the **self-similarity** of substitution tilings, the small-k Bragg-peak regime (where oscillations dominate) can be truncated, leaving a clean power-law tail in $E(t)$. Results: Fibonacci $\alpha = 3.0000 \pm 0.000008$; Period-Doubling $\alpha = 1.0000 \pm 0.00001$; Penrose tiling (2D) $\alpha = 5.97 \pm 0.06$ (close to the Class I value $d + 1 = 3$ for a 2D tiling with the Pisot property). The paper confirms $\phi_2 = 0.35$ for all 1D chains studied and provides the most precise numerical validation of the eigenvalue formula (Öguz et al. 2019) to date.

2.7 Order Metrics and Comparative Rankings

Middlemas, Stillinger, Torquato (2019). This paper computes $\bar{\Lambda}$ for all Barlow sphere-packing sequences in $d = 3$, parametrized by fcc/hcp stacking codes. A linear model $\bar{\Lambda}(\alpha) = \alpha\bar{\Lambda}_{\text{fcc}} + (1 - \alpha)\bar{\Lambda}_{\text{hcp}}$ holds for periodic stacking sequences (where α is the fraction of fcc layers). The

dominant factor in $\bar{\Lambda}$ is the local cluster geometry: fcc-like environments (60° tetrahedral) are more hyperuniform than hcp-like environments (90° tetrahedral). Theta-series methods are used for all Bragg-peak sums, establishing the Zachary–Torquato (2009) technique as standard. The paper demonstrates that $\bar{\Lambda}$ serves as a sensitive discriminator among structures that share the same pair-correlation function to short range.

Morse, Steinhardt, Torquato (2024). This paper studies ordered versus disordered stealthy hyperuniform systems across spatial dimensions $d = 1$–4. In $d = 1$ the densest Fourier-space hard-sphere packing — which saturates the stealthy constraint with the fewest degrees of freedom — corresponds exactly to a Bravais lattice. In higher dimensions, disordered stealthy configurations can achieve *lower* packing density than their crystalline counterparts, counter to the usual expectation. The **order metric** τ (based on pair-distance statistics) is used as a complement to $\bar{\Lambda}$: τ is large for ordered (lattice-like) stealthy configurations and small for disordered ones. The results motivate using both $\bar{\Lambda}$ and τ to fully characterize stealthy hyperuniform materials, and demonstrate that stealth alone does not determine the degree of order.

2.8 Published Reference Values

The table below collects all α and $\bar{\Lambda}$ values that appear explicitly in the published literature, restricted to the $d = 1$ patterns directly relevant to this project plus the 2D Penrose tiling. Values marked † were confirmed independently by this project.

Pattern	**Class**	α **(published)**	$\bar{\Lambda}$ **(published)**	**Source**
Integer lattice ($d = 1$)	I	∞	1/6 †	Torquato & Stillinger (2003)
Fibonacci chain	I	3 (exact) †	0.20110 †	Zachary & Torquato (2009); Öguz et al. (2019); Hitin-Bialus et al.
Period-Doubling chain	II	1 (exact) †	diverges	Öguz et al. (2019); Hitin-Bialus et al.
Bombieri-Taylor chain	I	1.545 †	—	Bombieri & Taylor (1986); Öguz et al
URL ($a > 0$, $d = 1$)	I	2 †	$(1 + a^2)/6$ †	Klatt, Kim, Torquato (2020)
URL cloaked ($a = 1$)	I	2 †	1/3 †	Klatt, Kim, Torquato (2020)
Stealthy (d=1, various χ)	I	n/a †	see PRX Table V	Torquato, Zhang, Stillinger (2015)
Penrose tiling ($d = 2$)	I	5.97 ± 0.06	—	Hitin-Bialus et al. (2024)

† Independently confirmed by numerical computation in this project. "—" indicates the value was not reported in the cited work. $\bar{\Lambda}$ values for the Period-Doubling and Bombieri-Taylor chains were *not* published prior to this project. The Penrose tiling α is consistent with $d + \alpha_{1\mathrm{D}} = 2 + 3 = 5$ but the precise interpretation in 2D remains open.

3 Week 1

3.1 Code Benchmarking and Validation

The core numerical algorithm for computing $\sigma^2(R)$ was validated against the 1D Poisson process ($\sigma^2 = 2\rho R$) to within **1.1% mean relative error** across the full R-range. Output: `fig1_poisson_benchmark.png`.

3.2 Quasicrystal Generation

Two generation methods were implemented:

1. **Substitution method** (`substitution_tilings.py`): iterative rule application up to $N \sim 10^7$ for all five metallic-mean chains.

2. **Cut-and-project method** (`projection_method.py`): Fibonacci chain from $\mathbb{Z}^2$ projection. Agreement with substitution: $< 0.1\%$ on $\bar{\Lambda}$.

Chain	**Metallic mean** μ_n	$\lvert\det M\rvert$	α (exact)
Fibonacci ($n = 1$)	$(1+\sqrt{5})/2 \approx 1.618$	1	3
Silver ($n = 2$)	$1+\sqrt{2} \approx 2.414$	1	3
Bronze ($n = 3$)	$(3+\sqrt{13})/2 \approx 3.303$	1	3
Copper ($n = 4$)	$2+\sqrt{5} \approx 4.236$	1	3
Nickel ($n = 5$)	$(5+\sqrt{29})/2 \approx 5.193$	1	3

All 2×2 metallic-mean matrices have $|\det M| = 1$, which forces $|\lambda_2| = 1/\lambda_1$ and thus $\alpha = 3$ exactly for all of them.

3.3 Number Variance and $\bar{\Lambda}$

$\sigma^2(R)$ was computed for all five metallic-mean chains at $N \sim 10^6$–10^7. All are Class I (bounded, oscillating variance). The exact formula for the integer lattice is $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ denotes the fractional part.

Pattern	$\bar{\Lambda}$	**Notes**
Integer Lattice	$1/6 = 0.1667$	Literature minimum
Fibonacci	0.2011 ± 0.001	Confirms Zachary & Torquato (2009)
Silver	0.2500	Novel (later confirmed $= 1/4$ exactly; see §4)
Bronze	0.2821 ± 0.0006	Novel
Copper	0.293	Novel
Nickel	0.310	Novel

3.4 Spreadability

Point patterns are embedded into two-phase media ($\phi_2 = 0.35$). The excess spreadability $E(t) = \mathcal{S}(\infty) - \mathcal{S}(t)$ yields α from the long-time slope $E(t) \sim t^{-(1+\alpha)/2}$.

Pattern	α (measured)	α (expected)	Error
Fibonacci	3.049	3	1.6%
Silver	2.992	3	0.3%
Bronze	2.987	3	0.4%

Integer lattice confirmed $\alpha \to \infty$ (exponential decay). Spreadability was validated as a second, independent measure of α.

4 Week 2

4.1 Uncorrelated Random Lattice and Stable Distributions

The uncorrelated random lattice (URL) displaces each site by $U(-a, a)$. The exact formula $\bar{\Lambda}(a) = (1 + a^2)/6$ was derived and validated to $< 0.3\%$ for all a. URL gives $\alpha = 2$ for all $a > 0$ (Class I). At $a = 1$ (cloaking limit), $\bar{\Lambda} = 1/3$ exactly — this turns out to be the limit of the metallic-mean sequence.

Symmetric α-stable perturbations fill the $1 < \alpha < 2$ space stochastically (Class II/III for stability index $s < 1$).

4.2 Stealthy Hyperuniform Patterns

Stealthy configurations ($S(k) = 0$ for $|k| < K^*$) were analyzed:

- Collaborator-provided ensemble: ∼4300 disordered-phase configurations per χ ($N = 2000$, $\rho = 1$, $\chi \in \{0.1, 0.2, 0.3\}$).
- 1D approximation derived: $\bar{\Lambda} \approx 1/(\pi^2\chi)$, validated below.

χ	$\bar{\Lambda}$ (measured)	$\bar{\Lambda} = 1/(\pi^2\chi)$	Excess
0.10	1.021	1.013	+0.8%
0.20	0.526	0.507	+3.7%
0.30	0.357	0.338	+5.6%

Two-phase discovery: Generating stealthy configurations via collective-coordinate optimization from a *perturbed lattice* starting point gives $\bar{\Lambda} \approx 1/6$ — near the lattice minimum — while collaborator data (random starting positions) gives $\bar{\Lambda} \approx 1/(\pi^2\chi) \gg 1/6$. Both phases satisfy $S(k) = 0$ for $|k| < K^*$ but differ drastically in their spatial structure. This shows stealthy hyperuniformity does not uniquely determine a point pattern.

Re-classification: Stealthy patterns are not power-law suppressed; α is not meaningful. The correct descriptor is $K^* = 4\pi\rho\chi$.

4.3 New Substitution Chains

Three new chains were added to `substitution_tilings.py`:

Chain	**Rules**	**Matrix**	**Class**	α
Period-Doubling	$S \to L$, $L \to LSS$	$[[0,2],[1,1]]$	II	1.000
0222 Chain	$S \to LL$, $L \to SSLL$	$[[0,2],[2,2]]$	III	0.639
Bombieri-Taylor	$a \to aac$, $b \to ac$, $c \to b$	3×3	I	1.545

Bombieri-Taylor fills the $1 < \alpha < 2$ gap:

$$M_{BT} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad \alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}, \quad \bar{\Lambda} = 0.377$$

This is the first known Class I quasicrystal with $1 < \alpha < 2$.

A new $\alpha = 2$ chain was also added (first deterministic substitution tiling at this value, previously only the stochastic URL occupied $\alpha = 2$):

$$a \to bc,\ b \to c,\ c \to abc \quad \Rightarrow \quad \lambda_1 \approx 2.148,\ |\lambda_{2,3}| \approx 0.682 \text{ (complex pair)},\ \alpha = 2.000,\ \bar{\Lambda} = 0.275.$$

4.4 Full Catalog Assembly and Junior Paper

All patterns were compiled into `catalog.json` via `research_catalog.py`. Infrastructure: `disordered_patterns.p` `stealthy_analysis.py`, `make_figures.py`.

The Junior Paper (`jp/jp_hyperuniformity.tex`) was drafted: §1 Introduction, §2 Theoretical Background, §3 Methods (5 subsections). 9 pages, biblatex+biber backend. Results sections (§4–§8) exist as a separate draft (`jp/jp_results_draft.tex`) pending organization.

5 Week 3

Work completed between the second and third meetings.

5.1 Spreadability Convergence for Bombieri-Taylor

Figure: `fig_bt_deep.png`

Spreadability converges to $\alpha = 1.545$ for BT, but very slowly: $\hat{\alpha} = 2.26$ at $t \in [10^2, 10^5]$, 1.98 at $[10^4, 10^7]$, 1.51 at $[10^8, 10^{11}]$. Two causes:

1. Large $|\lambda_2| = 0.802$ (vs. 0.618 for Fibonacci) means correction terms in the spreadability decay slowly.
2. BT has 24× more Bragg peaks than Fibonacci (10^5 vs. 4,300), causing oscillations that bias slope fits.

$\bar{\Lambda}$ from $\sigma^2(R)$ avoids these issues (no fitting window needed); $\bar{\Lambda}_{\mathrm{BT}} = 0.377$.

5.2 Finite-Size Convergence

Figure: `fig_finite_size_alpha.png`

$\bar{\Lambda}$ from $\sigma^2(R)$ is stable from $N \sim 10^3$ for both BT and Fibonacci. BT spreadability $\hat{\alpha}$ converges from ∼2.4 toward 1.545 but needs $t > 10^8$. Fibonacci spreadability works at moderate t.

5.3 3×3 Substitution Matrix Search

Figure: `fig_new_tilings.png`

Enumerated all 39,304 3×3 non-negative integer matrices with $|\det M| = 1$ and row sums ≤ 4. Of these, 4,882 satisfy $|\lambda_i| < 1$ for $i \geq 2$. Among those, 504 have complex conjugate eigenvalue pairs; all 504 give $\alpha = 2$ exactly (18 distinct λ_1 values, range 1.32–3.85). The remaining 4,378 matrices (all-real eigenvalues) give $\alpha = 3$. No matrix gives $\alpha \in (2, 3)$.

The $\alpha = 2$ result follows from the determinant constraint: for 3×3 with $|\det M| = 1$ and complex pair $\lambda_2 = \bar{\lambda}_3$,

$$\lambda_1 |\lambda_2|^2 = 1 \;\Rightarrow\; |\lambda_2| = \lambda_1^{-1/2} \;\Rightarrow\; \alpha = 2 \text{ exactly.}$$

Example: $a \to bc$, $b \to c$, $c \to abc$ ($\lambda_1 \approx 2.148$, $\bar{\Lambda} = 0.274$). $\bar{\Lambda}$ varies widely across the 504 matrices (0.26 to 1.7), even among those sharing the same λ_1.

5.4 Generalized Metrics for Classes II and III

Figure: `fig_generalized_metrics.png`

$\bar{\Lambda}$ is not defined for Classes II and III ($\sigma^2(R)$ grows without bound). Instead:

$$\Lambda_{II} = \lim_{R \to \infty} \frac{\sigma^2(R)}{\ln R} \qquad \text{(Class II)}$$

$$\Lambda_{III} = \lim_{R \to \infty} \frac{\sigma^2(R)}{R^{1-\alpha}} \qquad \text{(Class III)}$$

Extracted via curve fit: $\sigma^2 = \Lambda_{II} \ln R + b$ (Class II) and $\sigma^2 = \Lambda_{III} R^{1-\alpha} + b$ (Class III, α fixed to theory). Results: Period-Doubling $\Lambda_{II} = 0.082$; 0222 chain $\Lambda_{III} = 0.143$ (with $\alpha = 0.639$ from eigenvalue formula).

5.5 The $2 < \alpha < 3$ Gap

Figure: `fig_nonunimodular.png`

Extended the 3×3 search to 4×4: 24M matrices tested, 1.36M with $|\lambda_i| < 1$, zero with $\alpha \in (2, 3)$. The gap persists because:

- Rank 3 with complex pair: $|\det M| = 1$ forces $\alpha = 2$ (proof above)
- Rank ≥ 4 with complex pair: extra small eigenvalues force $|\lambda_2|$ larger, giving $\alpha < 2$
- All-real eigenvalues: numerical search finds only $\alpha = 3$

No substitution matrix with $|\det M| = 1$ at any rank can produce $\alpha \in (2, 3)$.

Non-unit determinant: For 2×2 matrices with $|\lambda_2| < 1$:

$$\alpha = 3 - \frac{2 \ln |\det M|}{\ln \lambda_1}$$

To get $\alpha \in (2, 3)$: need $1 < |\det M| < \sqrt{\lambda_1}$. Since det is an integer, only $|\det M| = 2$ works (requires $\lambda_1 > 4$). Found 28 such matrices, all with $|\det M| = 2$ and $\alpha \in (2.0, 2.2)$.

Smallest example: $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$), rules $a \to b$, $b \to aabbbb$. $\lambda_1 = 2 + \sqrt{6} \approx 4.449$, $\alpha = 2.071$. $\sigma^2(R)$ oscillates over $[0.001, 0.87]$ (Fibonacci: $[0.15, 0.25]$). $\bar{\Lambda} \approx 0.648 \pm 0.004$ ($N = 17.8$M, period-averaged). First known 1D Class I chain with $\alpha \in (2, 3)$.

5.6 Silver $\bar{\Lambda} = 1/4$ and Metallic-Mean Convergence

Figures: `fig_silver_lambda_quarter.png`, `fig_metallic_convergence.png`

Silver: $N = 9.37 \times 10^6$ points, 5×10^4 windows: $\bar{\Lambda} = 0.2502 \pm 0.0006$ (0.3σ from $1/4$). Silver has $\rho = 1/2$ exactly. Fibonacci: 0.201 (ours) vs. 0.20110 (Zachary & Torquato 2009, theta-series; both numerical). Bronze: 0.2822 ± 0.0006 (no obvious closed form).

Metallic-mean sequence: Computed $\bar{\Lambda}(\mu_n)$ for $n = 1$ through 20:

n	1	2	3	4	5	6	7	8	10
$\bar{\Lambda}$	0.201	0.250	0.282	0.293	0.310	0.318	0.321	0.325	0.327
n	15	20	∞						
$\bar{\Lambda}$	0.329	0.331	1/3						

Monotone from below. Limit $= 1/3 = \bar{\Lambda}$ of URL cloaked lattice ($a = 1$, Klatt et al. 2020). As $n \to \infty$, $L/S = \mu_n \to \infty$ but $\ell_L / \langle \ell \rangle \to 1$: the long tile becomes the mean spacing, short tiles become rare and tiny, and the chain approaches a lattice with disorder matching URL $a = 1$.

6 Week 4

Tasks from the third meeting (2026-03-27) with Prof. Torquato.

6.1 Why $\alpha_{\max} = 3$ for 1D Substitution Tilings

The eigenvalue formula $\alpha = 1 - 2 \ln |\lambda_2| / \ln \lambda_1$ (Öguz et al. 2019) gives $\alpha = 3$ for all metallic-mean chains. The bound $\alpha \leq 3$ holds for *all* unimodular ($|\det M| = \pm 1$) substitution matrices, for the following reason.

For a rank-r substitution matrix M with Perron–Frobenius eigenvalue λ_1 and remaining eigenvalues $\lambda_2, \ldots, \lambda_r$, the determinant constraint gives

$$|\det M| = \lambda_1 \prod_{i=2}^{r} |\lambda_i| \geq \lambda_1 |\lambda_2|^{r-1} \quad \Rightarrow \quad |\lambda_2| \leq \left(\frac{|\det M|}{\lambda_1} \right)^{1/(r-1)}.$$

Substituting into the eigenvalue formula:

$$\alpha = 1 - \frac{2\ln|\lambda_2|}{\ln\lambda_1} \leq 1 + \frac{2}{r-1}\left(\frac{\ln\lambda_1 - \ln|\det M|}{\ln\lambda_1}\right) = 1 + \frac{2}{r-1}\left(1 - \frac{\ln|\det M|}{\ln\lambda_1}\right).$$

For $|\det M| = 1$ this simplifies to $\alpha \leq 1 + 2/(r-1)$. The bound is saturated when all sub-dominant eigenvalues have the same absolute value, i.e., $|\lambda_i| = \lambda_1^{-1/(r-1)}$ for $i \geq 2$.

Rank r	$\alpha_{\max}$	**Example**
2	3	all metallic means (Fibonacci, Silver, ...)
3	2	504 cubic matrices with complex pair
4	$5/3 \approx 1.67$	(no known examples saturating this)
$r \to \infty$	$\to 1^+$	approaches Class II boundary

Thus $\alpha = 3$ is the global maximum for unimodular substitution tilings in 1D, achieved uniquely by rank-2 (two-letter) chains. Higher-rank matrices necessarily have $\alpha < 3$ because the extra eigenvalues in $|\det M| = 1$ force $|\lambda_2|$ to be larger (closer to 1), reducing α. Non-unimodular matrices can exceed $\alpha = 3$ in principle (e.g. $|\det M| = 0$ gives $\alpha = \infty$, but this degenerates to a lattice), though in practice the range $\alpha \in (2, 3)$ requires $1 < |\det M| < \sqrt{\lambda_1}$ as shown in §3.5.

6.2 Physical Interpretation of the $\bar{\Lambda}$ Ranking

$\bar{\Lambda}$ quantifies residual large-scale density fluctuations: larger $\bar{\Lambda}$ means more disorder. The catalog ranking (§5) is controlled by a few structural parameters:

1. **Perfect order → minimum.** The integer lattice achieves $\bar{\Lambda} = 1/6$, the global minimum for all 1D hyperuniform patterns. Any departure from perfect periodicity increases $\bar{\Lambda}$.
2. **Metallic means: tile disparity controls disorder.** As n grows, the tile-length ratio $\mu_n = L/S$ increases. Short tiles become rarer (frequency $\sim 1/\mu_n \to 0$) and act as "defects" in an otherwise lattice-like array of long tiles. More extreme defects → larger variance → larger $\bar{\Lambda}$. The sequence $\bar{\Lambda}(\mu_1) = 0.201 < \bar{\Lambda}(\mu_2) = 0.250 < \cdots$ is monotone increasing toward 1/3.
3. **URL: displacement amplitude a controls disorder directly.** Each site is displaced by $U(-a/2, a/2)$, giving $\bar{\Lambda} = (1 + a^2)/6$. At $a = 0$: lattice ($\bar{\Lambda} = 1/6$). At $a = 1$: cloaked ($\bar{\Lambda} = 1/3$), matching the metallic-mean limit. Intermediate a interpolates smoothly.
4. **Stealthy: weaker constraint → more disorder.** The stealth condition $S(k) = 0$ for $|k| < K^*$ constrains fewer degrees of freedom when χ is small. $\bar{\Lambda} \approx 1/(\pi^2\chi)$: smaller χ allows larger fluctuations.
5. **Bombieri-Taylor: slow decay of $S(k)$ means large fluctuations.** $\alpha = 1.545$ is low (close to Class II boundary), so $S(k)$ decays slowly near $k = 0$, allowing more long-wavelength fluctuations. Combined with $|\lambda_2| = 0.802$ (close to 1), giving $\bar{\Lambda} = 0.377$.

In summary, $\bar{\Lambda}$ increases with: (i) larger tile-length disparity in substitution chains, (ii) larger displacement amplitude in perturbed lattices, (iii) weaker spectral constraints in stealthy patterns, and (iv) slower power-law suppression of $S(k)$ near the origin.

6.3 Uncloaked URL Patterns

Figure: `fig_url_sweep.png`

The exact formula $\bar{\Lambda}(a) = (1+a^2)/6$ (Klatt et al. 2020) was validated by simulation at 17 values of $a \in [0.05, 2.0]$ with $N = 50{,}000$ points each. All simulated values match the formula to within 1%.

a	$\bar{\Lambda}$ (exact)	$\bar{\Lambda}$ (sim)	Error	Note
0	0.1667	—	—	lattice
0.5	0.2083	0.2082	0.08%	
1.0	0.3333	0.3330	0.10%	cloaked (Bragg peaks vanish)
1.5	0.5046	0.5047	0.02%	
2.0	0.6667	0.6657	0.14%	second cloaking event

Key observations: (i) $\bar{\Lambda}$ grows monotonically with a (more displacement = more disorder). (ii) At each integer $a = 1, 2, \ldots$, cloaking occurs: the Bragg peaks of the underlying lattice vanish because the displacement form factor $\tilde{f}(k) = \sin(\pi k a)/(\pi k a)$ has zeros at all non-zero reciprocal lattice vectors. (iii) $\bar{\Lambda}(a = 1) = 1/3$ matches the metallic-mean limit exactly.

6.4 g_2-Invariant Patterns

Torquato & Stillinger (2003, §V.B) define a **g_2-invariant process** as one where the pair correlation function $g_2(r)$ remains invariant as density increases, up to a terminal density ρ_c at which the system becomes hyperuniform ($S(0) = 0$).

Three 1D examples were investigated:

1. **Lattice-gas:** Tessellate $[0, L)$ into N intervals of length $D = 1$; place one point uniformly at random in each interval. $g_2(r) = r/D$ for $r \leq D$, $g_2 = 1$ for $r > D$. This is equivalent to URL with $a = 1$ (just a phase offset: $j + U(0, 1)$ vs. $j + U(-1/2, 1/2)$). $\bar{\Lambda} = 0.333 = 1/3$, confirming the URL result.
2. **Step-function g_2:** $g_2(r) = \Theta(r - D)$ at terminal density $\phi_c = 1/2^d$. For $d = 1$: $\phi_c = 1/2$, close-packed hard rods = lattice. $\bar{\Lambda} = 1/6$.
3. **Gaussian shuffled lattice:** Each lattice site displaced by $\mathcal{N}(0, \sigma^2)$. Hyperuniform for any $\sigma > 0$. Measured: $\bar{\Lambda} = 0.200$ ($\sigma = 0.1$), 0.250 ($\sigma = 0.2$), 0.342 ($\sigma = 0.3$). At $\sigma = 0.1$, $\bar{\Lambda}$ matches Fibonacci; at $\sigma = 0.2$, it matches Silver = 1/4.

The g_2-invariant processes at terminal density are all hyperuniform and fall naturally into the existing ranking: they are particular cases of perturbed lattices, with $\bar{\Lambda}$ determined by the displacement distribution. The lattice-gas is a special case of URL. The step+delta g_2 examples (Eq. 121–125 in the 2003 paper) are 3D constructions with no direct 1D analog beyond the cases already computed.

6.5 Metallic Chain Tile Visualizations

Figure: `fig_metallic_tiles_extended.png`

Figure shows 25 tiles of each metallic-mean chain for $n = 1, 2, 3, 4, 5, 6, 8, 10$. As n increases, short tiles (S) become rarer: the S fraction decreases from 36% at $n = 1$ (Fibonacci) to 8% at $n = 10$. For large n, the chain approaches a near-lattice of long tiles with rare, widely-spaced short-tile defects.

6.6 Metallic-Mean Convergence to $n = 100$

Extended $\bar{\Lambda}(\mu_n)$ from $n = 20$ to $n = 100$ (TARGET_N = 300k).

n	1	2	3	5	10	20	30	50	70	100	∞
$\bar{\Lambda}$	0.199	0.250	0.283	0.310	0.327	0.332	0.332	0.334	0.330	0.335	1/3

All values are consistent with 1/3 within statistical error at large n. Some nominal overshoots at $n \geq 50$ are within the ~ 0.005 error bars (reduced accuracy at large n because chains grow very slowly: $\lambda_1 \approx n + 1/n$, so even at $n = 100$ only two substitution iterations are needed to reach $N = 300$k, limiting the quasi-periodic structure). The overall picture is unchanged: $\bar{\Lambda}(\mu_n) \to 1/3$ **monotonically from below**, confirming the URL cloaked-lattice limit.

6.7 All Non-Unimodular Chains

Script: `compute_all_nonunimodular.py`

Enumerated all 2×2 non-negative integer matrices with $|\det M| = 2$, $|\lambda_2| < 1$, $\alpha \in (2, 3)$, and row sums ≤ 10. Found **138 matrices** (more than the original 28 found with row sums ≤ 6) across 12 distinct α values.

α	Count	$\bar{\Lambda}$ range	Smallest $\bar{\Lambda}$
2.071	12	$[0.32, 2.01]$	$M = \left[\begin{smallmatrix} 0 & 2 \\ 1 & 9 \end{smallmatrix}\right]$
2.087	10	$[0.34, 1.57]$	
2.175	18	$[0.33, 3.36]$	
2.199	14	$[0.34, 3.57]$	
2.248	10	$[0.33, 2.82]$	
2.271	16	$[0.34, 3.76]$	
2.301	18	$[0.33, 4.05]$	
2.343	12	$[0.33, 4.79]$	
2.362	12	$[0.34, 4.80]$	
2.376	8	$[0.33, 4.97]$	
2.393	4	$[0.34, 5.08]$	

Key observations: (i) All α values lie in $[2.07, 2.39]$, consistent with the formula $\alpha = 3 - 2\ln 2/\ln \lambda_1$ and the constraint $\lambda_1 > 4$. (ii) $\bar{\Lambda}$ varies enormously *within* each α group (by factors of 5–15), showing that α alone does not determine $\bar{\Lambda}$. (iii) The smallest $\bar{\Lambda} \approx 0.32$–$0.34$ across all groups, close to 1/3 (the URL cloaked value). (iv) These are all Class I ($|\lambda_2| < 1$, bounded σ^2).

6.8 Yuan-Torquato Improved Spreadability

Script: `yuan_improved_spreadability.py`

Implemented the improved fitting procedure from Yuan & Torquato (2026, arXiv:2602.17873). Three fitting function types were tested on Fibonacci ($\alpha = 3$):

Chain	**Method**	$\hat{\alpha}$	$\lvert\hat{\alpha} - \alpha\rvert$	Error
Fibonacci ($\alpha = 3$)	Simple log-slope	2.19	0.81	27%
	Type I, order 1	3.04	0.04	1.3%
	Type III, order 1	2.95	0.05	1.5%
BT ($\alpha = 1.545$)	Simple log-slope	2.27	0.72	47%
	Type I, order 6	1.57	0.03	**1.7%**
	Type III, order 6	2.16	0.61	40%

For Fibonacci, Type I at low order ($n = 1$) gives the best result (1.3% error); higher orders diverge because quasicrystal $s^{ex}(t)$ oscillates (dense Bragg peaks). For BT, Type I at high order ($n = 5$–6) dramatically reduces the error from 47% to $\sim 2\%$ — the **first time BT's α has been measured accurately from spreadability without extreme time ranges** ($t > 10^8$ was previously needed). Type III fails for BT because the spectral density is not analytic at the origin.

Conclusion: The Yuan-Torquato Type I fitting is a significant improvement for chains with large $|\lambda_2|$ (like BT), where correction terms dominate at moderate t. It does not replace variance-based $\bar{\Lambda}$ extraction but provides a viable alternative for α measurement from spreadability data.

7 Complete Pattern Catalog

Patterns sorted by $\bar{\Lambda}$ within Class I.

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	$1/6 = 0.167$
URL ($a = 0.1$)	I	2.0	0.168
URL ($a = 0.3$)	I	2.0	0.181
Gaussian $\sigma = 0.1$	I	2.0	0.187
Fibonacci	I	3.0	0.201
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	$\mathbf{1/4} = 0.250$
Cubic α=2 (new)	I	2.0	0.275
Bronze	I	3.0	0.282
Copper	I	3.0	0.293
Nickel	I	3.0	0.310
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	—†	0.358
Bombieri-Taylor	I	1.545	0.377
Stable $s = 1.7$	I	1.7	0.242
Stable $s = 1.5$	I	1.5	0.303
Stable $s = 1.3$	I	1.3	0.417
Stealthy ($\chi = 0.2$)	I	—†	0.526
Stealthy ($\chi = 0.1$)	I	—†	1.021
Period-Doubling	II	1.0	div. ($\Lambda_{II} = 0.080$)
0222 Chain	III	0.639	div. ($\Lambda_{III} = 0.209$)
Cauchy $\gamma = 0.1$	II	0.95	div.
Stable $s = 0.7$	III	0.7	div.
Stable $s = 0.5$	III	0.5	div.
Stable $s = 0.3$	III	0.3	div.

† Stealthy: $S(k) = 0$ by construction; α not applicable. Green rows: literature-confirmed benchmarks. Yellow: novel additions.

8 Figures and Scripts

Figures

Filename	Content
`fig1_poisson_benchmark.png`	Poisson variance validation
`fig2_bounded_variance.png`	$\sigma^2(R)$ for Fibonacci/Silver/Bronze
`fig3_hyperuniformity_test.png`	Normalized variance across classes
`fig9_stealthy_sk.png`	$S(k)$ for stealthy at $\chi = 0.1, 0.2, 0.3$
`stealthy_Sk_overlay.png`	$S(k)$ overlay of all stealthy configs
`bombieri_taylor_analysis.png`	BT variance, $S(k)$, spreadability
`fig12_ranking.png`	Full $\bar{\Lambda}$ ranking, Class I
`fig_variance_II_III.png`	$\sigma^2(R)$ for Period-Doubling and 0222
`fig_bt_deep.png`	BT oscillations, Bragg peak density, $E(t)$
`fig_finite_size_alpha.png`	$\bar{\Lambda}(N)$, $\Lambda_{II}(N)$, $\hat{\alpha}(N)$ (portrait)
`fig_finite_size_alpha_landscape.png`	Same, landscape 1×4 for slides
`fig_new_tilings.png`	Cubic α=2 chain variance and $S(k)$
`fig_generalized_metrics.png`	Normalized variance plateaus (Class II/III)
`fig_finite_size_lambda.png`	Lattice $\bar{\Lambda}$ analytic vs. simulated
`fig_catalog_updated.png`	Updated $\bar{\Lambda}$ ranking with stealthy fixes
`fig_metallic_convergence.png`	$\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$ converging to 1/3
`fig_4x4_alpha_dist.png`	4×4 matrix search: α distribution
`fig_nonunimodular.png`	$\sigma^2(R)$ and running $\bar{\Lambda}$ for $\alpha = 2.071$ chain

Scripts

Script	Purpose
`substitution_tilings.py`	All chains (metallic means, BT, cubic α=2, non-unimodular)
`quasicrystal_variance.py`	$\sigma^2(R)$, $\bar{\Lambda}$, generalized metrics
`projection_method.py`	Cut-and-project Fibonacci chain
`two_phase_media.py`	Spectral density, spreadability
`disordered_patterns.py`	URL, stealthy generation, exact $\bar{\Lambda}$ for URL
`stealthy_analysis.py`	Collaborator stealthy ensemble analysis
`research_catalog.py`	Full pipeline $\rightarrow$ `catalog.json`
`make_figures.py`	Figures from catalog + stealthy results
`bt_deep_dive.py`	(Mtg 2–3) BT oscillations, $S(k)$ density, $E(t)$
`finite_size_scaling.py`	(Mtg 2–3) $\bar{\Lambda}(N)$, $\Lambda_{II}(N)$, $\hat{\alpha}(N)$
`search_cubic_tilings.py`	(Mtg 2–3) Exhaustive 3×3 matrix search
`generalized_metrics.py`	(Mtg 2–3) Λ_{II}, Λ_{III}; patches variance module
`finite_size_lambda_bar.py`	(Mtg 2–3) Lattice formula; stealthy two phases
`metallic_mean_convergence.py`	(Mtg 2–3) $\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$
`search_4x4_tilings.py`	(Mtg 2–3) 4×4 matrix search (24M matrices)
`silver_precision_analysis.py`	(Mtg 2–3) High-precision Silver/Bronze $\bar{\Lambda}$
`fig_nonunimodular_chain.py`	(Mtg 2–3) Figure for $\alpha = 2.071$ non-unimodular chain

Presentations and Written Work

Document	Location	Notes
Week 1 presentation	`results/week1/week1_presentation.tex`	Beamer, core metallic-mean results
Week 2 presentation	`results/week2/week2_presentation.tex`	Beamer, 12 slides
Week 3 presentation	`results/week3/week3_presentation.tex`	Beamer, 16 slides (rewritten 2026-03-24)
Junior Paper	`jp/jp_hyperuniformity.tex`	9 pages, §1–§3 complete
JP results draft	`jp/jp_results_draft.tex`	§4–§8 + appendix, held for later

All documents compile via MiKTeX (`C:/Users/minec/AppData/Local/Programs/MiKTeX/miktex/bin/x64/`).

9 Key Theoretical Results

1. $\alpha = 3$ **universality of metallic means.** For any 2×2 metallic-mean matrix with $|\det M| = 1$: $|\lambda_2| = 1/\lambda_1 \Rightarrow \alpha = 3$ universally.

2. **Bombieri-Taylor fills $1 < \alpha < 2$.** The 3-letter cubic substitution $a \to aac, b \to ac, c \to b$ gives $\alpha = 1.545$, the first known 1D quasicrystal with $\alpha \in (1, 2)$.

3. **First deterministic $\alpha = 2$ substitution tiling.** The chain $c \to abc, a \to bc, b \to c$ gives $\alpha = 2.000$ exactly, $\bar{\Lambda} = 0.275$.

4. **$\bar{\Lambda}$ density invariance.** $\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$; direct comparison across patterns with different densities is valid.

5. **Integer lattice has no finite-N correction.** $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$ (piecewise cubic integral identity).

6. **Stealthy two-phase structure.** Both ordered and disordered stealthy phases satisfy $S(k) = 0$ for $|k| < K^*$, but differ drastically in $\bar{\Lambda}$ ($\approx 1/6$ vs. $1/(\pi^2\chi)$). Stealthy α is not a power-law exponent.

7. **$2 < \alpha < 3$ gap is inaccessible to unimodular matrices.** Exhaustive search over 3×3 (39,304) and 4×4 (24M) matrices confirms zero unimodular cases with $\alpha \in (2, 3)$. Rank-3 proof: $|\det M| = 1 \Rightarrow \alpha = 2$ exactly for complex pairs. Rank-≥ 4 proof: additional Pisot factor forces $\alpha < 2$.

8. **General 2×2 formula.** For any 2×2 Pisot matrix: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. Gap $(2, 3)$ is accessible when $|\det M| \in (1, \sqrt{\lambda_1})$; all 28 known examples have $|\det M| = 2$.

9. **First 1D quasicrystal with $\alpha \in (2, 3)$.** $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$) gives $\alpha = 2.071$, $\bar{\Lambda} \approx 0.650$ (rough estimate). Large log-periodic oscillations in $\sigma^2(R)$, period $\ln\lambda_1 \approx 1.493$.

10. **Silver $\bar{\Lambda} = 1/4$ exact (numerical).** $N = 3.88 \times 10^6$, 3×10^4 windows: $\bar{\Lambda} = 0.2501 \pm 0.0004$ (0.3σ from $1/4$). Analytical proof unknown.

11. **$\bar{\Lambda}(\mu_n) \to 1/3$ confirmed.** Metallic-mean sequence $n = 1 \ldots 20$ converges to $1/3$ at rate $0.22/n^{1.43}$. URL cloaked lattice ($a = 1$, $\bar{\Lambda} = 1/3$) is the exact limit.

10 Open Questions

1. $2 < \alpha < 3$ **gap:** ***RESOLVED*** **(mechanism and first example).** Non-unimodular $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ gives $\alpha = 2.071$. Sub-question: what is the minimal-complexity chain in $(2, 3)$? (Row sum ≥ 6 is currently the minimum requirement.)

2. **Silver exact value:** ***RESOLVED numerically.*** $\bar{\Lambda} = 1/4$ at 0.3σ. Analytical proof not yet known.

3. $\bar{\Lambda}$ **metallic-mean limit:** ***RESOLVED.*** $\bar{\Lambda}(\mu_n) \to 1/3$; rate $0.22/n^{1.43}$; URL cloaked lattice is the exact limit.

4. **Non-unimodular** $\bar{\Lambda}$**:** Is $\bar{\Lambda} = 0.650$ exactly $13/20$? Analytical proof unknown; large oscillation amplitude makes high-precision measurement difficult.

5. **Bronze closed form:** Is $\bar{\Lambda}(\text{Bronze}) = 0.2822$ a clean expression? Closest simple fraction: $11/39 = 0.28205$ (0.18σ, non-obvious).

6. **Stealthy two-phase literature:** Is the ordered/disordered bifurcation discussed in existing stealthy ground-state literature?

7. **2D generalization:** Does the eigenvalue formula $\alpha = 1 - 2\ln|\lambda_2|/\ln\lambda_1$ extend to 2D substitution tilings (Penrose, Ammann-Beenker)?